MY CHIREMBA: APP FOR MEDICAL DIAGNOSIS

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Abstract

This document proposes the next generation technology that everyone has been waiting for, a novel system that provides medical diagnosis assistance to users. The system is called My Chiremba. The system is designed to help users understand their symptoms and possible causes by guiding them through a series of questions to help narrow down symptoms to a possible diagnosis. The document also entails the root problems that drove the innovation of the system, investigations undergone and the objectives and scope that My Chiremba will have to meet.

Introduction

The rapid pace of technological innovation has transformed nearly every aspect of our lives, revolutionizing the way we work, communicate, and even manage our health and wellbeing. In the field of healthcare, the integration of cutting-edge tools and techniques has ushered in a new era; of more efficient, effective, and personalized care; that promises to deliver more accurate diagnoses, targeted treatments, and preventive care tailored to the unique needs of each individual.

However, despite these strides, a stark reality persists. In many communities around the world, including Zimbabwe, a significant portion of the population continues to face daunting barriers when it comes to accessing quality healthcare services. Whether it's the prohibitive cost of visiting a doctor or the lack of essential medical equipment and diagnostic tools in underfunded healthcare facilities, these challenges have left far too many Zimbabweans with limited options for managing their health.

Evidence of the above claim is given by the research information below.

1. In an Al Jezeera article written on 28 April 2021 with the title *Rich or poor, in*

Zimbabwe crumbling healthcare is deadly for all, the researcher tells the story of a Chitungwiza resident called Leonard Mugwade, who accompanied his neighbour to Chitungwiza Central Hospital where they were turned away because the neighbour was said to have had a non-crucial case, but then died before the Leonard could get $25 to take the person to another private hospital.

2. In a Herald article written on 03 January 2020 with the title *public hospitals adjust charges*, research showed that as of that time the consultation fees for adults receiving treatment at Parirenyatwa Group of Hospitals are up from $15 to $200 for initial and $130 for subsequent visits. Children above five years would pay $100 as initial consultation and $60 for reviews. In light of this information Mr Rusike then concluded by saying, *the low-income population can no longer afford to get ill because they cannot afford expensive curative services*.

This unacceptable state of affairs underscores the urgent need for innovative solutions that address the systemic barriers to quality healthcare, ensuring that every individual, regardless of their financial situation or geographical location, can access the care they deserve.

Problem Statement

Many patients lack access to medical professionals who provide accurate and timely diagnosis and treatment. This can be due to financial difficulties or the patients staying in remote or underserved areas where there are no nearby clinics or hospitals. (**It is costly**)

Healthcare providers often face time constraints in conducting comprehensive assessments and consultations with patients. The limited time available can lead to delays in treatment and ultimately poorer health outcomes for clients. (**It takes too much time**)

Related Work

An article was published in the National Library of medicine website with the title *Chatbots and Conversational Agents in Mental Health: A review of the Psychiatric Landscape*. The results of the study showed that conversational agents had potential for benefit in psychoeducation and self-adhering and also satisfaction rating of chatbots were extremely high, therefore suggesting that it would be an effective and at the same time enjoyable tool in psychiatric treatment

Artificial Intelligence-Based Symptom Checker Systems: A Review and Critical Appraisal by Johnson et al. (2020): This review article explores the existing AI-based symptom checker systems and evaluates their accuracy, usability, and clinical relevance. It discusses the challenges associated with symptom analysis and provides insights into the potential of AI-driven systems for improving diagnosis and decision-making in healthcare.

Another article on with the title *Mobile Health Applications for Diagnosis and Management of Medical Conditions* by Lalloo et al. (2019) examines the use of mobile health (mHealth)

applications for diagnosis and management of medical conditions. It discusses the role of mHealth apps in symptom tracking, self-diagnosis, and remote monitoring of patients. The review also addresses the challenges and opportunities associated with mHealth app adoption in healthcare.

Viz.ai is a mobile app that uses artificial intelligence to detect diseases such as intracerebral haemorrhage and vessel occlusions. This app also provides mobile imaging and HIPPA-complaint messaging to help synchronize workflow across many specialties for diseases such as stroke.

EpicCare is an electronic record system that allows health professionals to update patient's records in real time and also allows for the integration of various medical devices and data sources.

Infermedica offers an AI-driven symptom checker that enables users to describe their symptoms. The system analyzes the input and provides potential diagnoses and recommendations for further medical evaluation

Buoy Health is a web based ai health assistant that provides personalized clinical support and helps people make the best decisions about their health. It gives people insight on what to do when they are sick

Solution

It is against this background that the My Chiremba project emerges as a beacon of hope, giving a digital health solution that leverages the power of artificial intelligence to bring the benefits of personalized healthcare guidance directly to the fingertips of users. The objectives are:

- To provide real-time interaction which uses natural language processing for semantic analysis of the description of symptoms
- To provide a functionality for diagnosing a user based on given symptom
- To provide a functionality for giving the users daily health advices

And the benefits are:

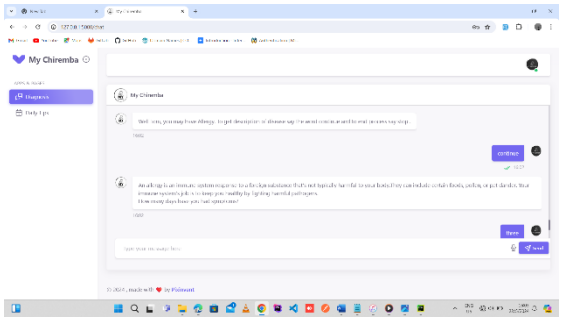
- Improved Healthcare Accessibility: Many people in Zimbabwe do not have access to healthcare facilities or cannot afford medical care. By leveraging the proposed technology, the My Chiremba can help bridge this gap by providing healthcare assistance to individuals who may have limited access to medical professionals or healthcare facilities, particularly in remote or underserved areas.

- Early Disease Detection: The timely identification and diagnosis of diseases significantly impact patient outcomes. My Chiremba can serve as an efficient screening tool, enabling individuals to receive preliminary assessments and recommendations promptly, leading to early detection and

intervention. With this the app can also help reduce the burden on healthcare facilities by providing pre-primary care for quick guidance so that red flags are timely detected by doctors and patients

- Personalized Healthcare: The app can utilize artificial intelligence algorithms to analyze individual symptoms, medical history, and other relevant factors to provide personalized insights and recommendations to users, empowering them to make informed decisions and take proactive measures for their health.

Results and Future Works



The system provided proper diagnosis and further provided complementary information to better understand and address the disease being faced

Objectives	Fully achieved	Partially achieved
To provide real-time interaction which uses natural language processing for semantic analysis of the description of symptoms	✓	
To provide a functionality for diagnosing a user based on given symptom	✓	
To provide a functionality for giving the users daily health advices	✓	

Future Works

1. Medical Image Analysis: The system will be able to diagnose a user based on the picture of the skin disease he would have uploaded on the system
2. Multilingual assistance: The system should allow communication even using languages like Shona and Ndebele
3. Nearest Doctor's Assistance: When the system would have determined the criticality of the disease, it should be able to recommend a specific doctor well vest with that particular disease and the doctor should have an account on the system

Conclusion

As the healthcare landscape continues to evolve, My Chiremba stands poised to play a pivotal role in empowering patients, reducing unnecessary medical costs, and fostering a more engaged and informed population. By democratizing access to medical expertise and putting the power of personalized healthcare guidance in the hands of users, the project seeks to redefine the way individuals manage their wellbeing and interact with the healthcare system.

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It is through the collective efforts of all the participants above that drove to the successful completion of the My Chiremba 2024 capstone project.

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